Effective Central Charges across Clean, Disordered, and Monitored Criticality: Benchmarks and an Exploratory Learning-Induced Metal–Insulator Transition

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Universal finite-size terms provide stringent tests of numerical calculations at two-dimensional critical points, but disorder and monitored dynamics complicate both their interpretation and their uncertainty. We establish a reproducible validation hierarchy using three benchmarks and then apply it to an exploratory learning-induced metal–insulator transition. For the clean Ising model we obtain $c = 0.498739$ from Monte Carlo data and $c = 0.499424$ from the exact transfer matrix. For the random-bond Ising model on the Nishimori line and a weakly self-dual disordered model we find effective central charges $c_{\mathrm{eff}} = 0.456469$ and 0.444107, respectively. At the candidate monitored transition $\phi/\pi = 0.30$, entanglement and Casimir fits instead yield 3.060740 and 12.579933. Their strong disagreement, together with an unbracketed transition and unstable anisotropy calibration, makes the monitored central-charge inference inconclusive. The comparison shows how validated benchmarks, resampling at the correct independence level, and explicit claim gates prevent precise-looking fits from being mistaken for universal results.

I. INTRODUCTION

Continuous phase transitions are distinguished by the loss of a microscopic length scale. If a control parameter t measures distance from criticality, the correlation length diverges as

$$\xi(t) \sim \xi_0 |t|^{-\nu}. \tag{1}$$

At $t = 0$, fluctuations occur on all scales between the lattice spacing and the system size. The renormalization group turns this observation into a calculus: coarse graining generates a flow in the space of Hamiltonians, critical theories are fixed points of that flow, and perturbations are classified as relevant, irrelevant, or marginal according to whether they grow, shrink, or remain finite under rescaling [1]. Microscopic systems attracted to the same fixed point share critical exponents and scaling functions and therefore form a universality class. This is why a lattice magnet, a fermion network, and a monitored quantum circuit can meaningfully be compared even though their microscopic variables are entirely different.

Two-dimensional criticality is unusually constrained. Under standard assumptions, scale invariance is enhanced to conformal invariance, and local conformal transformations generate the infinite-dimensional Virasoro algebra. A conformal field theory is characterized not only by scaling dimensions and operator-product coefficients but also by the central charge c, which appears in the anomalous term of the stress-tensor algebra [2, 3]. The central charge is sometimes described as a measure of low-energy degrees of freedom, but its most operational meaning for numerical work is as the coefficient of the conformal anomaly. Mapping the plane to a cylinder changes the vacuum energy by an amount fixed by this anomaly. Consequently, a periodic cylinder of circumference L has a universal Casimir correction proportional to c/L^2 [4–6]. A subsystem of length ℓ on a periodic quantum chain has a logarithmic entanglement entropy with the same coefficient [7]. These two observables provide independent routes from raw numerical data to a dimensionless universal number.

The coefficient is attractive precisely because it is restrictive. A critical temperature or coupling depends on microscopic conventions, whereas c should not. Reproducing $c = 1/2$ for the two-dimensional Ising fixed point, whose thermodynamics is exactly known [8], tests far more than the visual linearity of a graph. It simultaneously tests the normalization of the partition function, the sign of the free energy, the factor relating a strip Lyapunov exponent to an areal density, the cylinder boundary conditions, and the finite-size extrapolation. In Monte Carlo work it also tests the integration of an energy observable from a known reference point and the treatment of serial correlation. Cluster algorithms such as the Wolff update suppress critical slowing down [9], but they do not remove the obligation to identify independent streams and to propagate their uncertainty through the nonlinear fit.

The apparent economy of the central charge can therefore be deceptive. The universal term is subleading to the bulk free energy, and an error that is small on the scale of the total free energy may be large on the scale of its L^{-2} coefficient. Corrections from irrelevant operators can imitate a change in slope over a short size range. Measurements made along one Markov chain are correlated, so their row count is not their effective sample size. Quadrature errors in thermodynamic integration are systematic rather than statistical. Finally, choosing a fit window after seeing which one agrees with the target converts a validation calculation into a selection-biased one. A credible result thus needs exact or microscopic oracles, nested discretizations, independent stochastic replicas, residual diagnostics, and predeclared size-window tests. The pur-

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pose of our clean benchmark is to exercise this entire chain before any disorder-specific interpretation is introduced.

Quenched randomness changes both the fixed point and the statistical object. The Harris criterion predicts when spatial fluctuations of a local critical coupling destabilize a clean fixed point [10]; subsequent field theory shows how random fixed points and logarithmic corrections emerge in two-dimensional Potts and Ising systems [11]. More basically, one must distinguish quenched and annealed averages. If a bond realization J is static on the equilibration time scale, the physical free energy is $[\ln Z_J]_{\rm dis}$, whereas annealed disorder gives $\ln[Z_J]_{\rm dis}$. These operations are not interchangeable. The replica method represents the quenched logarithm by a derivative or $n \to 0$ limit of $[Z^n]_{\rm dis}$, so different replica sectors can carry different effective central charges. In nonunitary or disordered theories the finite-size coefficient is accordingly denoted $c_{\rm eff}$: it is a well-defined universal amplitude for a specified averaged observable, not automatically the central charge of a unitary Hilbert-space theory.

The random-bond Ising model on the Nishimori line is an especially stringent test. A relation between temperature and the bond distribution produces a gauge identity that fixes otherwise nontrivial disorder averages [12]. Duality and gauge structure locate a multicritical region [13], while transfer-matrix, supersymmetric, and network formulations expose its random fixed point [14, 15]. Numerical estimates near $c_{\rm eff} = 0.464$ provide a reference for the ordinary quenched free energy [16]. This qualifier is essential. A Born-weighted trajectory ensemble or a higher-replica moment need not measure the same logarithm and may legitimately yield another coefficient. In addition, widths obtained from the same disorder realization are correlated. A hierarchical or paired bootstrap must resample the realization as a whole; resampling widths or transfer rows as if independent would manufacture precision. Bootstrap resampling is useful here not because it eliminates modeling choices, but because it carries the empirically observed covariance through the complete fitting procedure [17].

Disordered fermions add a complementary viewpoint. The Chalker–Coddington construction expresses localization as propagation through a network of scattering nodes [18], and free-fermion mappings connect related networks to random-bond Ising transfer matrices [15, 19]. Anderson transitions separate localized and extended single-particle states; their universality depends on dimension and on the symmetry constraints of the Hamiltonian [20]. The Altland–Zirnbauer classification organizes those constraints by time-reversal, particle-hole, and chiral symmetries [21]. Its tenfold-way formulation classifies topological insulators and superconductors [22–24]. Symmetry class DIII, relevant below, has time-reversal symmetry squaring to -1 together with particle-hole symmetry. A DIII metal–insulator transition can therefore have scaling behavior that cannot be inferred by simply importing the clean Ising value.

Our third benchmark lies between the conventional quenched magnet and the open monitored problem. It propagates a fermionic Gaussian state while sampling outcomes from the state-conditioned Born rule. Gaussian covariance matrices permit efficient simulation of fermionic linear optics [25], but efficiency alone is not validation. Sequential Born sampling correlates outcomes through the evolving state; roundoff can also drive a covariance matrix away from the fermionic purity and antisymmetry manifold. Weak self-duality supplies an independent physical audit: electric and magnetic event densities must agree within their joint uncertainty even though individual trajectories do not obey a pointwise equality. We combine this test with Gaussian invariants, effective sample size, residual trends, and fit-window perturbations. The resulting coefficient is again an effective central charge of a precisely specified weighted ensemble.

Measurement-induced phenomena broaden the problem further. Unitary dynamics normally spreads entanglement, whereas local measurements remove quantum information. Competition between these processes can produce an entanglement transition between volume-law and area-law trajectory phases [26, 27]. Purification dynamics gives a related operational diagnostic [28]; mappings of random circuits to statistical mechanics explain how measurement outcomes generate replica interactions and nonunitary critical theories [29, 30]. Numerical studies find nontrivial critical scaling in Clifford and Haar-random circuits [31, 32], while monitored free fermions can display extended critical regimes and distinct routes to an area law [33]. Thus "measurement-induced" names a mechanism, not one universal fixed point.

The learning-induced model considered here combines trajectory weighting, free-fermion evolution, topology, and localization. A control angle ϕ changes the learned or decohering channel, and a transfer diagnostic is used to search for a class-DIII metal–insulator transition [34]. Three logical steps must remain separate. First, the transition must be localized: data must identify both sides of a crossing rather than merely a monotone trend. Second, the critical observable must be mapped to a conformal geometry. A lattice transfer direction and a spatial cut generally have different metric factors, so an anisotropy parameter plays the role of a velocity. Third, universal estimators should agree after this calibration. Entanglement and Casimir amplitudes depend on different raw measurements; agreement is therefore a powerful test of the assumed critical point and observable mapping.

These requirements motivate a distinction between parameter uncertainty and claim uncertainty. A regression can return a narrow conditional interval for an amplitude at a chosen ϕ, yet the scientific statement "this is the universal central charge at the transition" can remain unsupported if ϕ_c is not bracketed or the anisotropy drifts. Increasing Monte Carlo steps attacks variance at fixed model, size, and parameter. It does not generate the missing side of a phase boundary, enlarge the accessible finite-size range, or correct a biased conversion factor. We therefore use a predeclared claim gate: a universal interpretation is released only if phase localization, calibration stability, and independent-estimator agreement all pass. Failed gates retain the numerical amplitudes as diagnostics while preventing a precise-looking but physically underdetermined number from being published.

Here we organize four frozen calculations into a single ascending validation hierarchy. The clean square-lattice Ising model tests normalization, Monte Carlo thermodynamic integration, and the finite-size coefficient against an exact transfer matrix. The Nishimori random-bond Ising model tests ordinary quenched averaging, gauge identities, and realization-level resampling. A weakly self-dual Born ensemble tests Gaussian propagation, state-conditioned sampling, self-duality, and common-disorder comparisons. Only after those benchmarks do we analyze the exploratory learning-induced DIII candidate. The same philosophy is applied throughout: validate the microscopic generator independently of the asymptotic fit, preserve the true independence unit in uncertainty estimation, perturb the finite-size model, and separate a stable computation from an allowed physical claim. Figure 1 summarizes the resulting hierarchy.

The article has two complementary outcomes. Quantitatively, all three benchmarks reproduce their reference values within bootstrap uncertainty, providing end-to-end evidence for the numerical mappings used in each ensemble. Methodologically, the monitored calculation demonstrates why benchmark success must not be transferred automatically to a new observable. Its putative DIII transition is not bracketed by the available scan, the anisotropy calibration is unstable, and the entanglement and Casimir conversions disagree. We report both amplitudes because they diagnose what the present calculation measures, but we do not average them or label either one a universal central charge.

A useful way to formalize this distinction is to divide the error budget into four layers. Sampling error is the fluctuation remaining when the ensemble, geometry, parameter, and estimator are fixed; it is the part conventionally summarized by a standard error. Discretization and finite-size error arise because numerical integration uses a finite grid and simulations use finite L; these errors are probed by nested grids, correction terms, and size cuts. Calibration error enters when a measured lattice amplitude must be multiplied by a velocity or anisotropy factor. Model-selection error enters when the candidate critical point, scaling window, or replica observable is not uniquely identified. These layers respond to different computational remedies. Longer runs reduce the first, denser grids and larger widths address the second, dedicated geometries constrain the third, and broader phase scans or additional observables resolve the fourth. Reporting only the covariance matrix of the final regression silently treats the other three as zero.

The hierarchy also clarifies the role of positive and negative controls. An exact positive control asks whether a known universal coefficient can be recovered with the same conventions used later. A microscopic control asks whether the stochastic generator samples the intended distribution, for example through the Nishimori energy identity or a Born-rule probability. A negative control asks whether an analysis that should not show the target effect remains null. None of these controls proves the final universality class by itself. Instead, each removes a family of alternative explanations: normalization mistakes, incorrect disorder frequencies, state-update drift, or underestimated serial correlation. This layered logic is more demanding than obtaining an acceptable χ^2, but it is also more informative when a calculation fails.

Reproducibility is treated here as part of the physics rather than as an afterthought. Random streams are fixed independently of thread scheduling; the generator, processed tables, and summaries are linked by immutable paths and hashes; plotted points are read from those numeric artifacts rather than from manually edited graphics; and headline values in the manuscript are generated from validated adapters. A reader should therefore be able to trace a quoted interval back to its resampling unit and a withheld claim back to the Boolean diagnostics that blocked it. This provenance does not make a finite-size ansatz correct, but it makes assumptions visible and prevents a later presentation step from changing the statistical object.

The remainder of the paper is organized as follows. Section II defines the four ensembles and their finite-size observables. Section III explains the Rust simulation kernels, Xoshiro256++ stream construction, Python analysis, correlation-aware resampling, and claim gates. Section IV presents the three validated central-charge benchmarks. Section V analyzes the learning-induced candidate and traces each failed inference gate to a specific diagnostic. Section VI compares error mechanisms and identifies the next decisive calculations. The appendices derive the observable mappings, specify the bootstrap, and provide a machine-readable audit of the withheld claim.

II. MODELS AND UNIVERSAL OBSERVABLES

A. Clean and random-bond Ising models

For Ising spins $s_i = \pm 1$ on a square lattice, the reduced Hamiltonian is

$$-\beta H = \sum_{\langle ij \rangle} K_{ij} s_i s_j. \tag{2}$$

The clean benchmark uses $K_{ij} = K_c = \frac{1}{2}\ln(1+\sqrt{2})$, whereas the quenched benchmarks draw bond variables once per disorder realization and hold them fixed during thermal sampling. On the $\pm J$ Nishimori line, the bond distribution and inverse temperature obey the Nishimori condition [12]. The weakly self-dual ensemble uses the

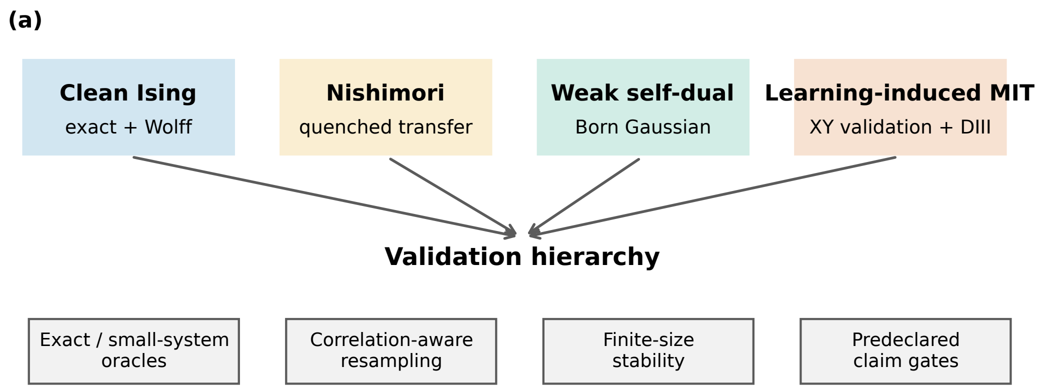

FIG. 1. Validation hierarchy used throughout this work. Exact and clean benchmarks validate normalization and scaling; disordered benchmarks add quenched averaging and common-disorder controls; the monitored calculation adds transition localization, anisotropy calibration, and agreement between independent universal estimators.

self-dual parameter specified by the frozen production data.

For an infinitely long cylinder of circumference L, conformal invariance gives

$$f(L) = f_\infty - \frac{\pi c_{\rm eff}}{6L^2} + a_4 L^{-4} + \cdots , \tag{3}$$

where f is the dimensionless free-energy density. In a unitary clean model $c_{\rm eff} = c$; for a quenched random system it denotes the effective central charge extracted from the disorder-averaged free energy. The averaging order is essential: the quenched observable is proportional to $[\ln Z]_{\rm dis}$, not $\ln[Z]_{\rm dis}$.

B. Monitored network model

The learning-induced problem is represented by a disordered free-fermion network in symmetry class DIII, with a measurement or decoherence angle ϕ controlling the trajectory ensemble. Its topological phases and possible intervening metallic regime are diagnosed by transfer-matrix localization observables and by the scale dependence of entanglement [19, 21, 34]. We analyze the frozen scan near $\phi/\pi = 0.30$ rather than retuning the candidate after inspecting its central-charge fits.

Two scaling relations are used. A periodic one-dimensional cut of length L has

$$S(\ell, L) = \frac{c_{\rm eff}}{3} \ln\left[\frac{L}{\pi}\sin\left(\frac{\pi\ell}{L}\right)\right] + s_0 + \delta S, \tag{4}$$

while the cylinder free energy follows Eq. (3) after converting the lattice aspect ratio with an anisotropy factor α. Agreement of these two estimates is a necessary internal consistency test; it is not imposed as a fitting constraint.

III. NUMERICAL METHODS AND VALIDATION HIERARCHY

A. Monte Carlo implementation

The Ising simulations are implemented in Rust. Every pseudorandom stream uses Xoshiro256++ [35], with seeds assigned before parallel execution so that scheduling cannot alter a realization. Thermal updates use the Wolff single-cluster algorithm [9]; measurements are separated by completed cluster updates, and warmup samples are discarded before accumulation. Energy-like observables are formed with a Rao–Blackwell estimator whenever the conditional bond expectation is available. This reduces thermal variance without changing the quenched sampling unit.

The simulation executable writes plain numerical records and metadata. Python is used only for validation, data reduction, nonlinear fits, bootstrap resampling, and publication graphics. This separation keeps the high-throughput stochastic kernel in a compiled language while making the statistical transformations directly inspectable.

B. Statistical estimators

For each bootstrap replicate we resample independent units and repeat the full fit, rather than perturbing the final parameter covariance matrix. Clean thermal

TABLE I. The four calculations and the additional controls required as model complexity increases.

Model	Universal probe	Principal control
Clean Ising	Casimir term	Exact transfer matrix
Nishimori RBIM	Quenched Casimir term	Disorder bootstrap
Weak self-dual	Quenched Casimir term	Common disorder
Monitored DIII	Entanglement/Casimir	Claim gates

TABLE II. Notation used across the four calculations. A symbol has the same geometric meaning wherever possible, while the measured ensemble is stated explicitly when it changes.

Symbol	Meaning	Symbol	Meaning
L	transverse circumference or chain length	M	transfer or imaginary-time length
K	reduced Ising coupling	p	antiferromagnetic-bond probability
ϕ	monitoring/decoherence control angle	α	spatial–temporal anisotropy factor
$f(L)$	free-energy density on a cylinder	$\gamma_1(L)$	leading Lyapunov/free-energy rate
$S(\ell, L)$	interval entanglement entropy	c	unitary conformal central charge
c_{eff}	ensemble-specific effective charge	$\text{CI}_{95\%}$	central 95% bootstrap interval

streams are the independent units. For random models a whole disorder realization—including every width measured from it—is resampled as a block. This paired bootstrap [17] preserves cross-width correlations. When two nearby parameter values are compared, the same bond realization and random labels are reused: this common-disorder construction reduces the variance of their difference without pretending that the pair is independent.

The reported standard error is the standard deviation of the bootstrap parameter distribution, and the interval is its central 95% percentile interval. Fit-window variation, truncation of small sizes, and correction terms are treated as systematic checks rather than folded into a single regression error. For the exact clean calculation no stochastic error is assigned.

a. Independence and uncertainty levels. The phrase *independent sampling unit* refers to the object that can be redrawn without reusing stochastic history. It is a complete Rust stream for the clean calculation, a complete bond realization for the Nishimori calculation, and a complete state-conditioned trajectory stream for the Born ensemble. Thermal uncertainty describes fluctuations conditional on a fixed Hamiltonian or disorder sample. Quenched-disorder uncertainty describes the variation between independently drawn static environments. The latter cannot be estimated by extending the transfer direction of one environment: more rows refine its conditional Lyapunov exponent but do not add a new disorder draw.

b. Conditional variance reduction. Rao–Blackwellization replaces a noisy event indicator by its exact conditional expectation given the current state. Common-disorder comparisons similarly reuse the same random environment at two parameter values so that shared fluctuations cancel in their difference. Both procedures reduce variance without changing the target expectation. Their uncertainty calculations must nevertheless preserve the pairing; treating the two members as independent would discard the covariance that produces the gain.

C. Systematic errors and fit selection

We separate four systematic categories from the bootstrap variance. Finite-size truncation is tested by removing the smallest width and by adding the leading symmetry-allowed correction term. Thermodynamic-integration discretization is tested on nested coupling grids. Transition-location uncertainty is controlled by demanding a two-sided phase bracket rather than assigning an error bar to a monotone scan. Anisotropy uncertainty is assessed from alternative spatial and temporal windows, not only from the standard error of one regression. Fit windows and correction powers are specified before comparison with the reference value; their shifts are reported as diagnostics and are not silently absorbed into a statistical error bar.

These distinctions determine which additional calculation is useful. More Monte Carlo steps improve thermal precision and, up to autocorrelation, effective sample size. More independent streams improve uncertainty across stochastic histories. Larger system sizes constrain irrelevant corrections and the asymptotic limit. A wider parameter scan localizes a transition, and new aspect ratios calibrate anisotropy. Exchanging one remedy for another can make an error bar smaller without making the physical inference more accurate.

D. Predeclared claim gates

We distinguish a numerical fit from a physical claim. Benchmark values must be stable to the prescribed size cuts and statistically compatible with their reference targets. A monitored universal-central-charge claim additionally requires (i) a transition bracketed by independent phase diagnostics, (ii) a stable space–time anisotropy calibration, and (iii) mutually compatible entanglement and Casimir estimates. Failing a gate does not erase the computed result; it changes the allowed statement from a universal estimate to a diagnostic conditional on the assumed candidate point.

IV. BENCHMARK CENTRAL CHARGES

A. Clean Ising calibration

The clean calculation uses periodic $L \times M$ tori with $M = 8L$ and $L = 4, 6, \ldots, 20$. We obtain the free energy at K_c in two independent ways. The first evaluates the exact strip transfer matrix. The second integrates the measured energy from the known $K = 0$ normalization,

$$F(K_c) = -N \ln 2 + \int_0^{K_c} \langle H \rangle_K \, dK, \tag{5}$$

on a nested 129-point coupling grid. The shift between 65- and 129-point Simpson quadrature is 0.001955 in the final central charge, below the Monte Carlo standard error. The cluster streams pass first/second-half and between-replica tests with the predeclared $|z| < 4$ threshold.

Figure 2 shows that the stochastic and exact free energies share the same Casimir slope. The primary $L_{\min} = 6$ fit, including an L^{-4} correction, gives

$$\begin{aligned} c_{\mathrm{MC}} &= 0.498739 \pm 0.005225, \\ c_{\mathrm{exact}} &= 0.499424. \end{aligned} \tag{6}$$

The Monte Carlo 95% interval $[0.488719, 0.509064]$ contains $1/2$. Diagnostic windows give $c_{\mathrm{MC}} = 0.499053$ for $L_{\min} = 4$ and 0.490505 for $L_{\min} = 8$; their increasing uncertainty and mutual compatibility show why the central window is used without selecting it a posteriori.

B. Nishimori random-bond Ising model

The $\pm J$ ensemble uses antiferromagnetic-bond probability $p = 0.1092212$ and $K_N = \frac{1}{2} \ln[(1-p)/p] = 1.0493604763$. For each of eight independent replicas, a matrix-free transfer product is propagated through 4096 burn-in rows and 2 097 152 measured rows at widths $L = 4, 6, 8, 10, 12, 14$. Normalizing after every row prevents overflow and turns the accumulated logarithmic norms into the quenched free-energy density.

The joint-width fit and hierarchical bootstrap in Fig. 3 give

$$\begin{aligned} c_{\mathrm{eff}} &= 0.456469 \pm 0.008181, \\ \mathrm{CI}_{95\%} &= [0.440064, 0.472304]. \end{aligned} \tag{7}$$

This interval contains the reference value 0.464 [16]. Removing $L = 4$ shifts the center from 0.456484 to 0.461475; a paired bootstrap of the difference contains zero. Independent microscopic audits also pass: the measured negative-bond fraction has $z = -0.524$, and a centered coupling derivative differs from the exact Nishimori energy identity by only 7.46×10^{-5}.

It is important to state the replica convention. Our value tests the *ordinary quenched* random-bond Ising free energy and therefore the reference $c_{\mathrm{eff}} \simeq 0.464$. The often-quoted value near 0.522 belongs to a different Born-weighted or higher-replica observable in monitored constructions. The two numbers are not competing estimates of the same quantity; substituting the higher-replica target into this benchmark would change the ensemble being measured.

C. Weakly self-dual Born ensemble

The third benchmark evolves a fermionic Gaussian covariance matrix at $\theta = \pi/4$ and $\beta = \beta' = 0.8813735870$. Sequential, state-conditioned Born sampling generates correlated outcomes according to

$$P(s|\Gamma) = \frac{1 + s \tanh(\beta) \langle i\gamma_a\gamma_b \rangle_\Gamma}{2}. \tag{8}$$

The leading Lyapunov or Shannon-free-energy rate is fitted over $L = 6, 8, \ldots, 32$ as $\gamma_1(L) = f_\infty L - \pi c_{\mathrm{eff}}/(6L) + a/L^3$. This Gaussian-state representation is consistent with standard efficient fermionic simulation methods [25].

The result,

$$\begin{aligned} c_{\mathrm{eff}} &= 0.444107 \pm 0.004240, \\ \mathrm{CI}_{95\%} &= [0.435749, 0.452494], \end{aligned} \tag{9}$$

contains the reference 0.447. As shown in Fig. 4, the maximum studentized residual is 1.772, its correlation with $1/L$ is -0.056, the minimum effective sample size is 4517, and the electric–magnetic self-duality statistic is $z = -1.460$. Required fit variants span 0.00553, smaller than the declared 0.01 systematic tolerance.

The compact comparison in Fig. 5 makes the hierarchy visible: all three stochastic intervals cover their predeclared references, but the meaning of the coefficient changes from a unitary central charge to a quenched or Born-weighted effective central charge.

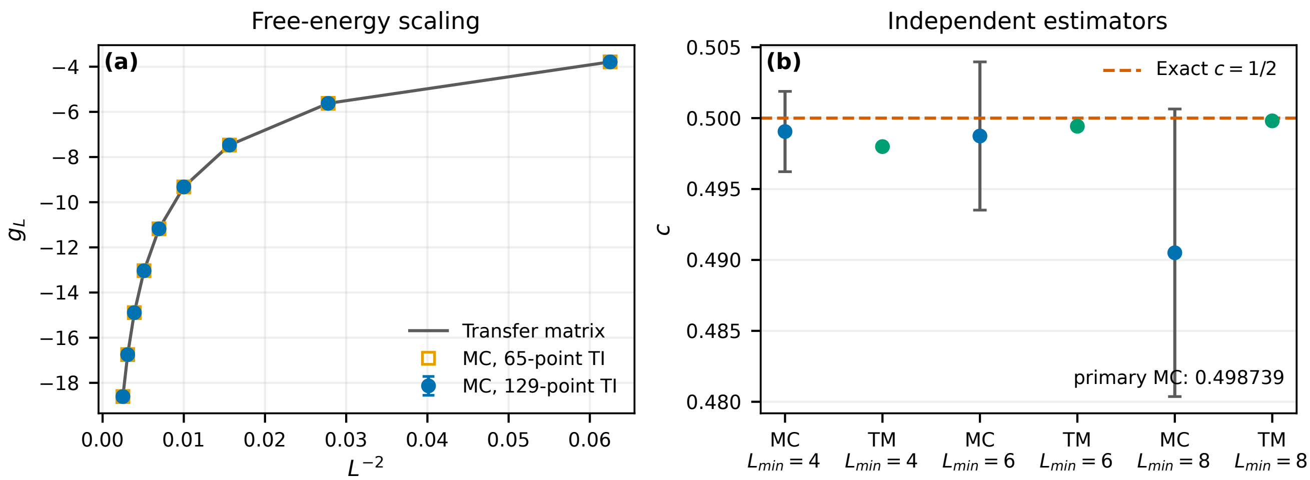

FIG. 2. Clean Ising calibration. (Left) Critical free-energy density versus L^{-2} from the exact transfer matrix and thermodynamic-integration Monte Carlo calculation. (Right) Independent estimates of c compared with the exact Ising value. Error bars are stream-level bootstrap uncertainties.

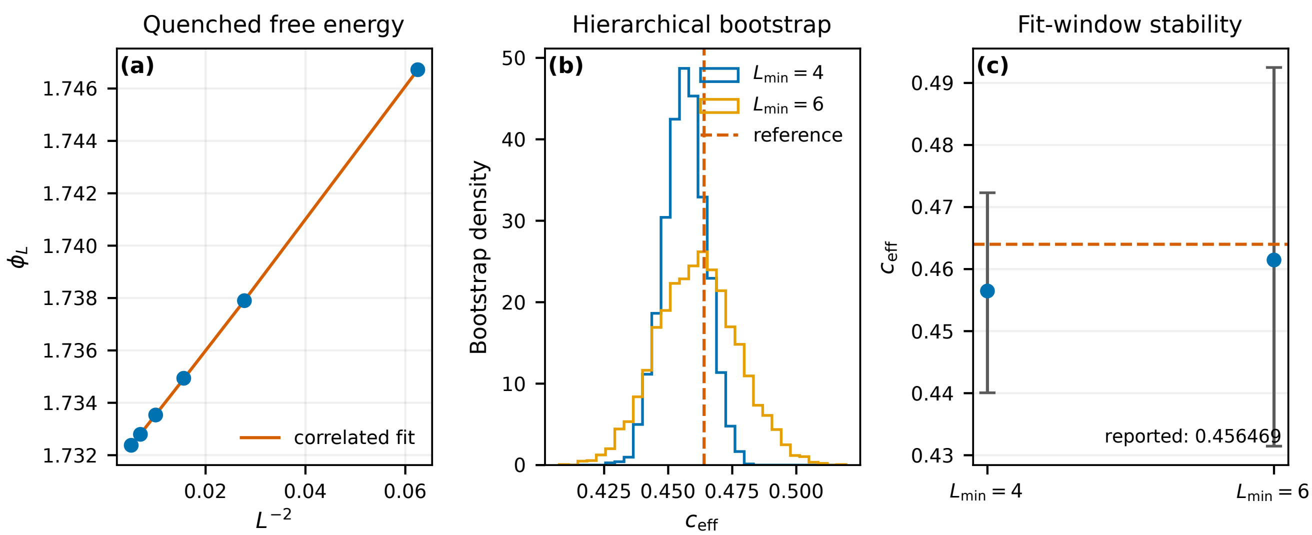

FIG. 3. Nishimori benchmark. (Left) Quenched free-energy density and the $L^{-2} + L^{-4}$ fit. (Center) Hierarchical bootstrap distribution of $c_{\rm eff}$. (Right) Stability under the lower-width cut. A disorder realization is resampled together at all widths.

V. EXPLORATORY LEARNING-INDUCED METAL–INSULATOR TRANSITION

A. Phase localization and candidate selection

Measurement-induced entanglement transitions can exhibit conformal scaling, but localization, trajectory weighting, and nonlinear replica limits all affect its interpretation [26, 27]. We therefore first validate the scanning observable in an XY slice. Its crossing lies in $\phi/\pi \in [0.24, 0.25]$, inside the reference window $[0.20, 0.28]$, as shown in Fig. 6. This confirms that the scan reacts correctly in the calibration geometry.

The generic DIII scan is different. Its diagnostic score increases monotonically from 0.023 at $\phi/\pi = 0.16$ to 1.387 at 0.32, but the available points do not exhibit both sides of a transition crossing. The transition is not bracketed. The boundary value $\phi/\pi = 0.30$ is retained as a preselected exploratory candidate, with the unobserved upper side extending to 0.32; it is not promoted to a critical point by the subsequent fits.

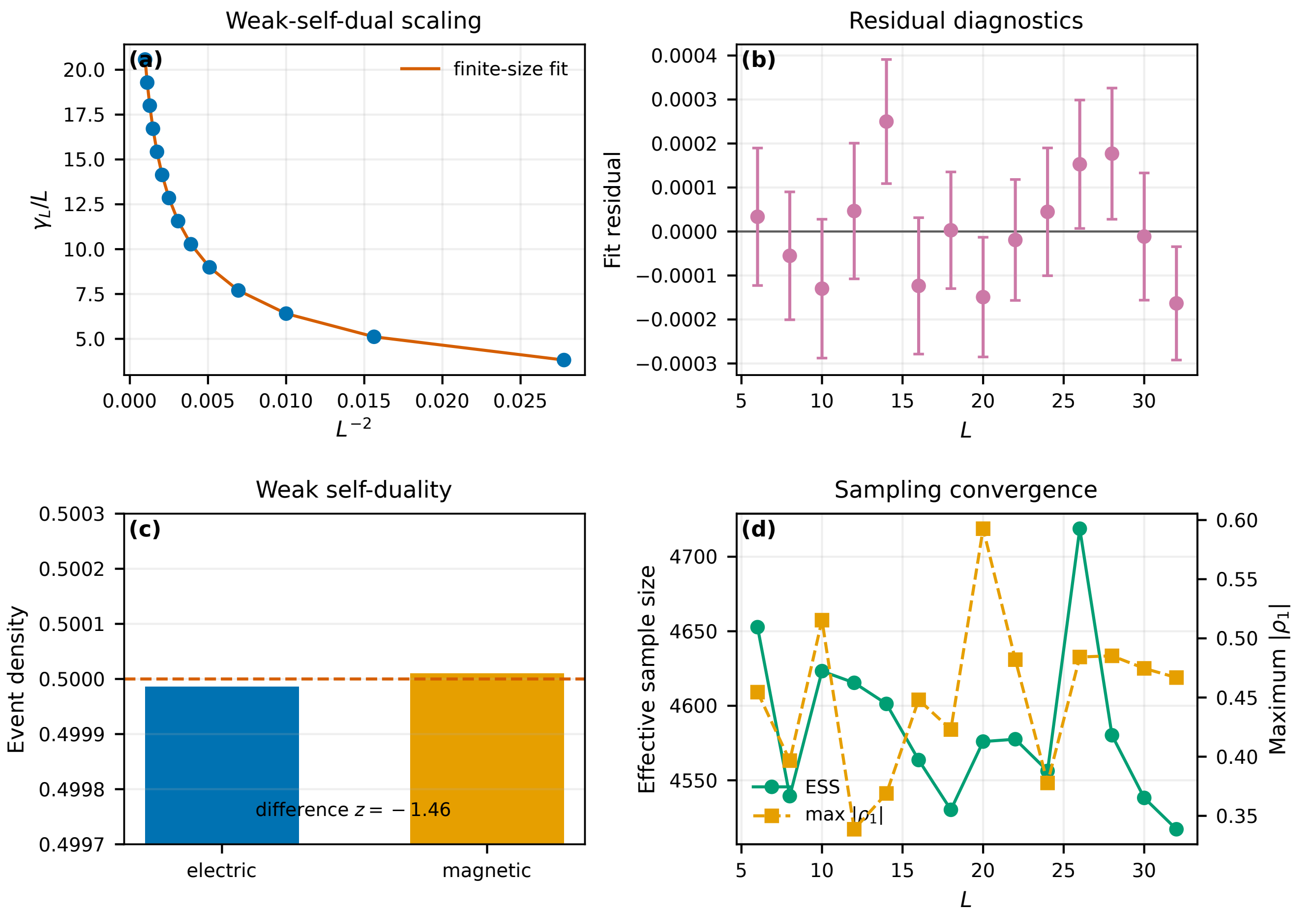

FIG. 4. Weak self-dual benchmark. The panels show the finite-size fit, studentized residuals, electric–magnetic self-duality audit, and autocorrelation/effective-sample-size convergence. The residual and self-duality diagnostics are evaluated independently of the target value.

TABLE III. Benchmark central charges. Parentheses denote one bootstrap standard error; intervals are central 95% bootstrap intervals. The exact clean result has no stochastic uncertainty.

Model	Widths	Estimate	95% interval	Reference	Outcome
Clean Ising, Monte Carlo	4–20	0.498739(0.005225)	[0.488719, 0.509064]	0.500	Pass
Clean Ising, exact	4–20	0.499424	—	0.500	Pass
Nishimori RBIM	4–14	0.456469(0.008181)	[0.440064, 0.472304]	0.464	Pass
Weak self-dual	6–32	0.444107(0.004240)	[0.435749, 0.452494]	0.447	Pass

B. Entanglement estimate

At each width $L = 8, 12, \ldots, 32$, the interval entropy is fitted to the chord coordinate in Eq. (4). The width-dependent coefficients are then extrapolated using a pre-specified model set; model weights are carried into the uncertainty rather than choosing the visually most favorable curve. Figure 7 shows both stages. The resulting conditional estimate is

$$c_{\rm eff}^{(S)} = 3.060740 \pm 0.759748,$$
$$\mathrm{CI}_{95\%} = [1.571634, 4.549845]. \tag{10}$$

The fit has $\chi^2/\mathrm{dof} = 0.370$ and remains available after removing the smallest width, but the covariance condition number is 6.9×10^4. The wide interval correctly reflects noisy per-width coefficients rather than the smoothness of a single $L = 32$ chord fit.

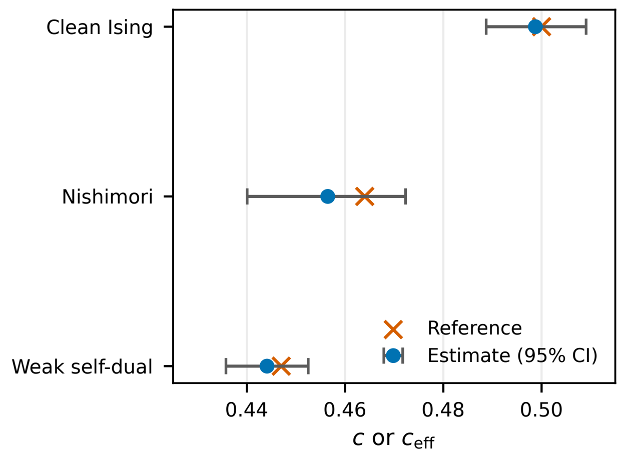

FIG. 5. Cross-benchmark comparison. Circles and horizontal bars show estimates and 95% intervals; crosses are reference values.

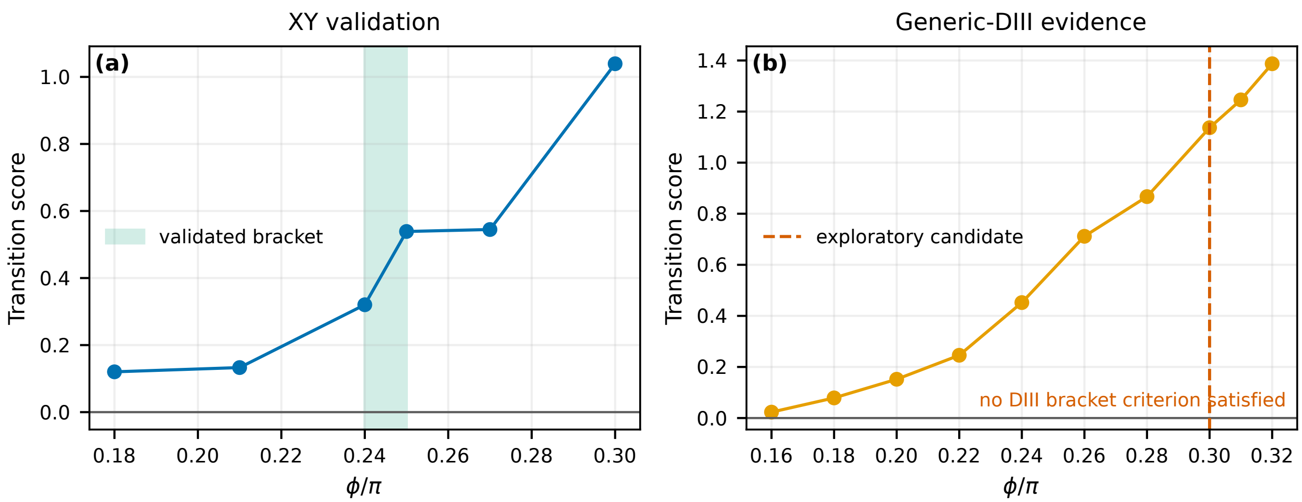

FIG. 6. Phase diagnostics. (Left) The XY validation scan brackets its transition between $\phi/\pi = 0.24$ and 0.25. (Right) The generic-DIII scan rises across the sampled interval but does not bracket a transition. The vertical marker at 0.30 is therefore an exploratory candidate rather than a localized critical point.

C. Casimir estimate and failed consistency gates

The free-energy amplitude can be converted to a central charge only after calibrating the spatial-to-temporal anisotropy. The frozen calculation gives $\alpha = 0.027675$ with a nominal interval $[0.02584, 0.02956]$. However, its relative spread across admissible windows is 2.69 and the spatial/temporal decay mismatch is 0.958. Thus the anisotropy calibration is unstable even though a single-window regression error is small.

Using the nominal α gives the conditional Casimir value

$$c_{\mathrm{eff}}^{(F)} = 12.579933 \pm 0.989806,$$
$$\mathrm{CI}_{95\%} = [10.727804, 14.610337]. \tag{11}$$

The difference $|c_{\mathrm{eff}}^{(F)} - c_{\mathrm{eff}}^{(S)}| = 9.519$ exceeds the combined 95% compatibility threshold 2.446: the estimators disagree. Figure 8 displays the apparently stable Casimir regression next to the unstable calibration and the final inconsistency. These diagnostics show why increasing Monte Carlo steps alone would not resolve the present problem. More steps reduce sampling variance, whereas the leading failures are an unbracketed parameter range, calibration drift, and estimator bias or mismatch.

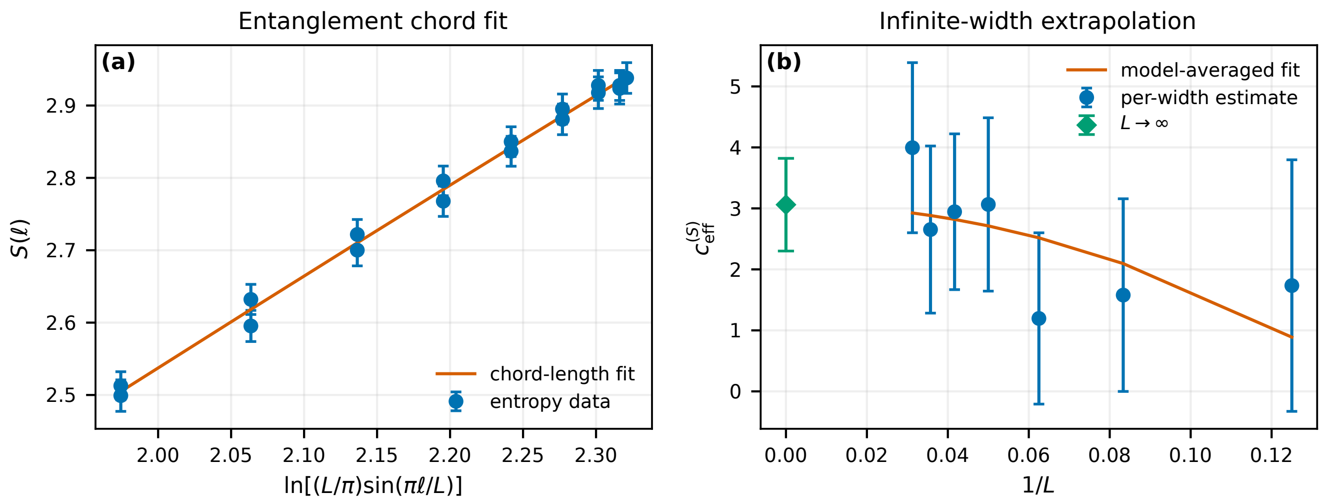

FIG. 7. Entanglement analysis at $\phi/\pi = 0.30$. (Left) Entropy versus conformal chord coordinate for the available widths. (Right) Extrapolation of the width-dependent effective coefficients. Model-selection uncertainty is included in the reported interval.

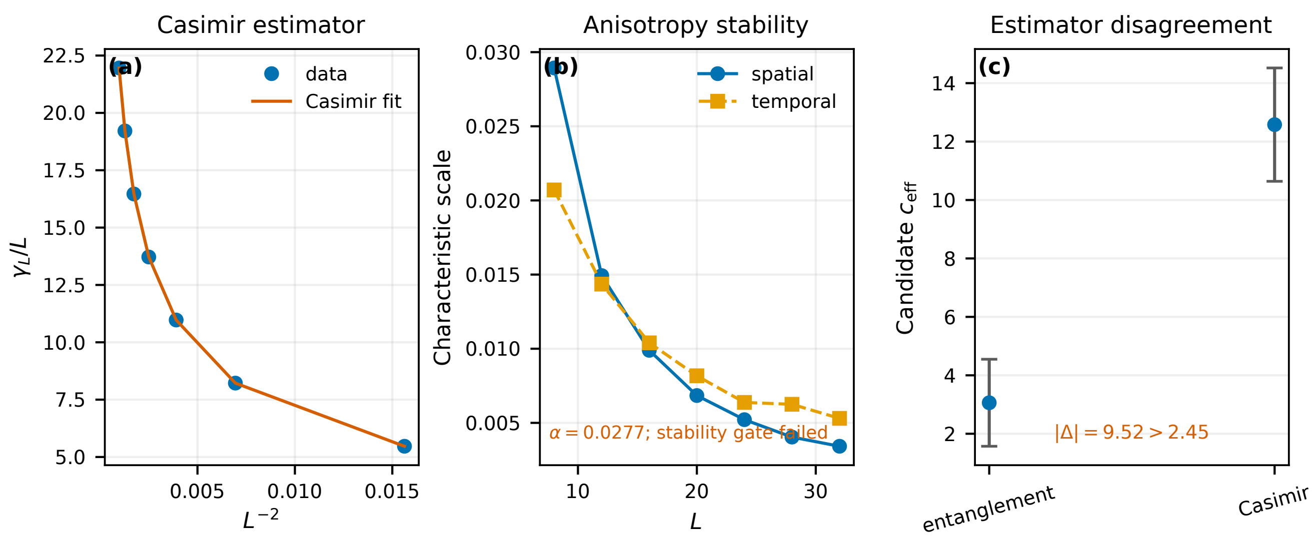

FIG. 8. Learning-induced-transition diagnostics. (Left) Casimir-amplitude fit at the exploratory candidate. (Center) anisotropy estimates under calibration variations. (Right) entanglement and nominally converted Casimir estimates, whose 95% intervals do not overlap.

TABLE IV. Conditional monitored-model results at $\phi/\pi = 0.30$. “No” means that the quantity is retained as a diagnostic but not published as a universal central charge.

Quantity	Value	Gate
XY bracket	$[0.24, 0.25]$	Pass
DIII bracket	None	Fail
$c_{\rm eff}^{(S)}$	3.060740(0.759748)	Available
α	0.027675	Unstable
$c_{\rm eff}^{(F)}$	12.579933(0.989806)	Available
Estimator agreement	No	Fail
Published central charge	false	Withheld

Consequently, this pair of numbers in Table IV does not constitute a universal-central-charge estimate. They are conditional diagnostics at an exploratory parameter value. This claim boundary is part of the result, not a qualification added after observing disagreement.

VI. CROSS-MODEL DISCUSSION

A. What the benchmark sequence validates

The common analysis separates three questions: whether a numerical observable is stable, whether its statistical error is correctly estimated, and whether the observable supports the intended universal interpretation. The clean model tests normalization, thermodynamic integration, the sign and factor of the Casimir coefficient, and finite-size corrections. Exact agreement rules out a broad class of coding errors that an internal goodness of fit cannot detect.

The Nishimori calculation then changes the statistical unit. Rows from one transfer product are correlated, and widths sharing a disorder realization are paired. Resampling rows or widths independently would underestimate the slope uncertainty. The identity and bond-frequency audits test the disorder generator separately from the fit. The weak self-dual model adds state-conditioned Born sampling: Gaussian invariant errors below 2.6×10^{-13} and the electric–magnetic check show that the measured coefficient is not merely a stable output of a drifting covariance matrix.

B. Error budget and parameter meaning

Table V distinguishes error sources that are often collapsed into one bar. Statistical uncertainty is reduced by additional independent streams or disorder realizations. Autocorrelation is reduced by longer runs only until the effective sample size becomes adequate. Finite-size bias requires larger L or controlled correction terms. Transition-location and anisotropy errors require new parameter coverage and calibration geometries; repeating the same point more accurately cannot remove them.

The large monitored values are therefore informative without being universal. They may reflect evaluation away from criticality, an incorrect anisotropic conversion, large finite-width corrections, or genuinely different replica observables. The present data do not discriminate among these mechanisms. In particular, the success of a smooth Casimir fit cannot substitute for agreement with entanglement or for a bracketed transition.

C. Next decisive calculation

The efficient next step is not uniform growth of the existing sample count. First, extend the DIII scan beyond 0.32 until the localization diagnostic shows a two-sided crossing, then refine that interval using common random numbers. Second, calibrate α from several aspect ratios at the bracketed point and require stability under width cuts. Third, recompute both universal estimators with larger widths and bootstrap their difference using shared disorder. Only if the difference interval contains zero should a combined central-charge estimate be quoted. This staged allocation directs computation at the dominant uncertainty rather than at the smallest error bar.

VII. CONCLUSION

We have placed four finite-size calculations in one PRB-style validation framework. Clean Ising thermodynamic integration yields $c = 0.498739(0.005225)$ and agrees with the independent exact value 0.499424. Realization-level resampling gives $c_{\mathrm{eff}} = 0.456469(0.008181)$ for the ordinary quenched Nishimori model, and state-conditioned Born sampling gives $c_{\mathrm{eff}} = 0.444107(0.004240)$ for the weakly self-dual benchmark. Each reference lies inside its 95% interval and independent implementation gates pass.

At the learning-induced metal–insulator-transition candidate, entanglement and nominal Casimir conversions produce 3.060740(0.759748) and 12.579933(0.989806), respectively. Because the DIII transition is unbracketed, the anisotropy is unstable, and the estimators are incompatible, false is correctly recorded as the publication status of a central charge. The defensible result is a reproducible failure of the universality gates together with a concrete refinement path. Across all four models, the central lesson is that more Monte Carlo steps improve accuracy only when sampling noise is the limiting error; they cannot repair an unlocalized critical point or an uncontrolled observable mapping.

Appendix A: Fit models and window selection

All fit forms follow from an extensive cylinder free energy plus the leading conformal Casimir term. For clean Ising and the Nishimori free-energy density we use

$$f(L) = f_\infty - \frac{\pi c_{\mathrm{eff}}}{6L^2} + \frac{a_4}{L^4}. \tag{A1}$$

Periodic translation-invariant geometry excludes an L^{-1} density term, and the L^{-4} contribution represents the leading retained irrelevant correction. The primary clean window begins at $L_{\min} = 6$; $L_{\min} = 4$ and 8 are diagnostic fits. The primary Nishimori window includes $L = 4$ because only six widths are available, while the paired $L_{\min} = 6$ fit measures sensitivity to that choice.

The weak-self-dual observable is stored as an extensive rate,

$$\gamma_1(L) = f_\infty L - \frac{\pi c_{\mathrm{eff}}}{6L} + \frac{a_3}{L^3}, \tag{A2}$$

which becomes the preceding density form after division by L. Fits that raise $L_{\min}$, double the block length, discard $L = 30$, or extend burn-in are paired with the primary data. The largest center shift and the total span

TABLE V. Error mechanisms, controls, and effective remedies.

Source	Diagnostic	Present control	Remedy if failed
Thermal variance	replica/half-chain z	Wolff streams, blocking	more independent clusters
Quenched variance	bootstrap spread	realization-level paired bootstrap	more disorder realizations
Autocorrelation	ESS	block analysis	longer streams or better updates
Finite-size bias	$L_{\min}$ drift/residuals	L^{-4} or L^{-3} terms	larger widths
Quadrature bias	nested-grid shift	65/129-point comparison	denser coupling grid
Transition location	phase bracket	XY positive control	extend/refine DIII scan
Anisotropy	window spread	spatial/temporal comparison	dedicated aspect-ratio calibration
Estimator interpretation	S–F difference	independent estimators	resolve mapping before universality claim

of required variants are compared with predeclared tolerances.

For monitored entanglement, a linear chord fit is performed separately at each L before the resulting coefficients are extrapolated. Keeping these two stages separate prevents the many interval points at one width from masquerading as independent evidence for the infinite-width limit. Alternative extrapolation powers receive fixed model weights. The Casimir fit uses the measured $\gamma_1(L)$ form, but conversion of its amplitude to c_{eff} is conditional on α. Hence a stable amplitude cannot override a failed anisotropy gate.

Window variation is a sensitivity analysis, not permission to select the answer nearest a reference. A variant is accepted when residual structure, precision, and predeclared stability criteria pass. Bootstrap replicates repeat the entire fit for the same window; paired replicates quantify the difference between windows. This procedure exposes both sampling dispersion and finite-size-model dependence while keeping their meanings distinct.

Appendix B: Observable mappings

For a periodic cylinder with transfer direction $M \gg L$, write $-\ln Z = Mg(L)$ and $f(L) = g(L)/L$. Conformal invariance gives

$$g(L) = f_\infty L - \frac{\pi c}{6L} + O(L^{-3}), \tag{B1}$$

which is Eq. (3) after division by L. This derivation fixes both the sign and the factor of six. The clean transfer matrix supplies an independent microscopic mapping to the same coefficient.

For the $\pm J$ model, $g(L)$ is obtained from the long-product Lyapunov exponent. The physical quenched object is

$$f_q(L) = -\lim_{M\to\infty} \frac{[\ln Z_{L,M}]_{\text{dis}}}{LM}. \tag{B2}$$

The sign convention of the stored logarithmic norm is converted once, before the finite-size fit. For the Born ensemble, $\gamma_1(L)$ is extensive in L, so the universal correction is L^{-1} rather than L^{-2} until the rate is divided by the width.

For monitored entanglement, the coefficient of the chord logarithm is first estimated separately at every width. The reported $c_{\text{eff}}^{(S)}$ is the large-L extrapolation of those coefficients. The Casimir conversion contains the independently calibrated α; therefore its uncertainty and stability are logically prior to interpreting $c_{\text{eff}}^{(F)}$.

Appendix C: Bootstrap construction

Let y_{rL} denote the estimate for realization or stream r at width L. A bootstrap replicate draws indices $r_1^*, \ldots, r_R^*$ with replacement and forms the complete vector

$$\bar{\boldsymbol{y}}^* = \frac{1}{R}\sum_{j=1}^{R} (y_{r_j^* L_1}, \ldots, y_{r_j^* L_n}). \tag{C1}$$

The nonlinear finite-size fit is repeated on this vector. All widths from one quenched realization consequently enter together, preserving the empirical cross-width covariance. The clean analysis applies the same construction to independent Monte Carlo streams. The Nishimori and weak-self-dual production analyses use 4000 replicates with fixed analysis seeds, but the simulation RNG streams remain independent of those seeds.

For common-disorder comparisons, parameter values a and b share the same bootstrap index in every replicate. The sampled quantity is $\Delta c^* = c_a^* - c_b^*$, not the difference of two independently resampled fits. This is the correct uncertainty for a paired fit-window or parameter comparison. Percentile intervals are read directly from the replicate distribution; no Gaussian shape is assumed.

Appendix D: Claim-gate audit

The frozen run contains 8 independent random streams and widths $8, 12, \ldots, 32$, with elapsed time 4583.689 s. Its summary artifact has SHA-256 digest `cc08a6e6d6d414046c744b4d29d48f112d44526dfc2145b867aae01f07d53c33`.

The claim gate evaluates three Boolean conditions. First, the DIII phase diagnostic must supply a lower and an upper boundary; it supplies neither a complete bracket nor a crossing in the available interval. Second, the anisotropy must be stable across prescribed windows; its relative spread is 2.69. Third, the independent 95% intervals for the entanglement and Casimir conversions must be compatible; their center difference 9.519 exceeds the combined threshold 2.446.

All three failures are recorded in the source summary as `diii_transition_not_bracketed`, `anisotropy_unstable`, and `estimator_disagreement`. No post-processing rule overrides them. The machine-readable publication flag is false. This audit ensures that a reader can reproduce not only the numerical values but also the logical reason they are not presented as a central charge.

1. Reproduction workflow

The analysis uses immutable result directories containing configuration, manifest, raw records, processed summaries, and plots. Rust executables generate stochastic records with Xoshiro256++ streams; Python scripts load hash-gated artifacts, perform resampling and fitting, and build the figures. The manuscript values are generated from those adapters rather than copied by hand. The complete workflow is invoked by the package Makefile, which runs tests, regenerates figures and TeX macros, compiles the RevTeX source, and verifies the resulting PDF.

DATA AVAILABILITY

All numerical results used in this article are retained in the accompanying repository as hash-identified result artifacts. Each artifact contains its configuration, manifest, raw or block-level records, processed summary, and analysis figures. No external proprietary data are required.

AUTHOR CONTRIBUTIONS

Xu Tian designed and implemented the numerical calculations, performed the analysis, and drafted the manuscript. Huidan Tan contributed to the physical interpretation, validation strategy, and revision of the manuscript. Both authors reviewed and approved the final paper.

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